

Preliminary Results for Novel Shear Force Sensor using Force Sensitive Resistors

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Abstract— This paper presents the theory, design, and development of an innovative shear sensor using force sensitive resistors (FSR) and two soft silicone layers that is intended to allow shear forces applied to skin to be measured. The sensor is being developed to better understand the shear forces applied to skin during patient transfers. The sensor is constructed using four FSRs arranged in a two by two matrix and secured between the two silicone layers. The silicone layers include features to secure the FSRs in place and concentrate the force applied to the sensor to the center of the FSR pads. The preliminary results show that the sensor allows the measurement of normal forces of up to 4.9 N applied to the 4.5cm square sensor to be measured with an R^2 of 0.996. The results also show that the sensor can measure shear forces of up to 1.5 N with an R^2 of 0.981. The results show the sensor has a linear response to both normal and shear forces. Future work is required to further analyze the performance of this novel sensor.

Keywords—*Shear Sensor, Force Sensitive Resistor, Shear Force Measurement*

I. INTRODUCTION

The process of transferring or lifting a patient that is mobility impaired can pose many challenges and potential risks for the patient and also for caregivers or health care professionals assisting the transfer. The transfer of patients using lift systems is demanding on staff resources as there needs to be a sufficient number of personnel and the amount of time needed to complete the transfer [1]. The most prominent staff risk involved in transfers are musculoskeletal risks such as injuries to the lower back, shoulders, wrist, and knees [2].

Transfers can also cause injuries to the patient involved. One of the most common injuries that occurs is pressure ulcers. This type of injury typically occurs on the soft tissues directly in contact with the lifting device. Pressure ulcers can either be caused by an abundance of force or a moderate amount of force over a sustained period [3] [4].

When moving a patient traditionally, a sling or lifting device is used and these methods for transfers predominantly apply normal forces to the patient equal to the patient's

weight concentrated to the lifting device such as a sling and shear forces to support limbs that are outside of the lifting device such as lower legs and feet. For analysis of how the normal force is distributed over a patient's body, it can be separated into 15 segments [5]. Most of the weight of a patient is distributed in three of these segments; the trunk, abdomen and pelvic area [6]. Traditional patient transfer devices are designed to support these segments which apply extra pressure as limbs such as legs are outside the lift, leading to risk to the core areas as they carry the full body weight which can lead to pressure ulcers [4].

As a person ages, their skin begins to lose its elasticity. The loss of elasticity reduces the total force needed to deform or damage and thus increases a patient's susceptibility for pressure ulcers [7]. The effect and risk of shear forces increases with these age-related changes although shear forces applied using conventional lift technology have received minimal study [4], [8].

There is a need for a device that can transfer a patient without causing injury and undue stress to the patients as well as the caregivers involved in the transfer. Able Innovations is developing a new device with an intent to transfer a patient from a bed without the application of any concentrated external forces to the patient. Specifically, the device does not use any lifting functions. To fully understand the shear forces experienced by the patient using conventional transfer lift technology and the proposed new system, this paper focuses on a new sensor design to allow the measurement of the forces that are exerted on a patient's skin including shear and normal forces.

Several methods exist to collect data on the forces that are acting on the body of a patient. Pressure mats are used to collect heat-maps of how the normal forces acting on the patient are distributed [6], [9]. These sensors are useful for gathering data where a force is applied in only one direction (normal) and specifically do not measure any shearing forces that may be occurring. Ultrasonic sensors and proximity sensors can determine where a person is located within a room as well as how they are moving [10],[11]. These sensors are not suitable for pressure measurement as they cannot measure the actual force that is being applied to the patient.

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Most of the above methods are also designed in such a way to provide real-time processing of data [12].

Applying a shear force to a soft body such as skin tissues causes the body to deform as it shifts and leans laterally because of the force [13]. The deformation is characterized by compression on one side of the material and decompression on the other side [14]. The basic principle of a shear sensor quantizes the difference in compression between both sides of a material using components that change their value as a function of how compressed they are [15]. An example sensor that can be used in shear sensors is pressure sensitive capacitors as their value changes with the inverse of the distance between the plates although this sensor requires a complex manufacturing process to imbed the capacitors within the simulated skin layer [13]. The directionality of the shear force can be determined by arranging the sensing elements in a grid and analyzing the pressure shift to identify the vector between the area of highest pressure increase to the area of highest pressure decrease within the array of sensors [16].

This paper will introduce an innovative sensor that is capable of measuring both shear and normal forces using force sensitive resistors FSRs, leather and molded silicone layers. The design of the shear sensor and the results of the characterization experiments performed on it are presented. Lastly, the advantages and disadvantages of this sensor will be discussed in addition to areas of further research.

II. DESIGN OVERVIEW

The transfer application assessment of shear requires that a sensor be able to be used as a proxy for the skin surface of a patient being transferred such as being applied to the outside of a manikin used during transfer tests. The result is that the sensor needs to be generally planar in design and that the sensor has similar surface characteristics to skin. Figure 1 shows the structure of the proposed sensor consisting of 4 layers.

The base theory of operation for the sensor is that a shear force applied to the skin surface (top) of the sensor will deform the silicone layers leading to a shift in the forces observed on the 4 sensors. For example, if a shear force to the left is applied, the two sensors (yellow/grey – left side) will see increased load and the two sensors (blue/brown – right side) will see a decreased load. A key assumption in the operation is the preloading of all the sensors with a normal force and this will occur as part of the application as the weight of the patient's body provides this force.

A. Hardware

The sensing component chosen for the sensor is four Interlink 402 Force Sensitive Resistors (FSR). These sensors can resolve forces from 0.1 N to 100 N and this is adequate for use in the shear sensor as the maximum force that could be applied to the sensor is 200N, therefore, the maximum force per FSR assuming even distribution is 50 N [17] across each of the 4 FSRs.

Figure 2 shows the basic circuit structure of the sensing system. The sensing structure of the sensor is a voltage

divider where the resistance variation for each FSR with the applied force leads to a change in the voltage from the respective voltage divider as an indication of the force applied. The known resistance (R_{known}) on the lower side of the voltage divider selects the range of forces that the sensor is sensitive to. A value of 10 k Ω was selected for R_{known} to allow for the output of the divider to cover the full range of the voltage rail from zero to five volts [17]. An Arduino Uno was used for the data acquisition unit in the sensor system because of its ease of use, suitability and availability in the lab. The Arduino transmits the values that it reads from the voltage dividers to a PC where the data is stored and subsequently processed.

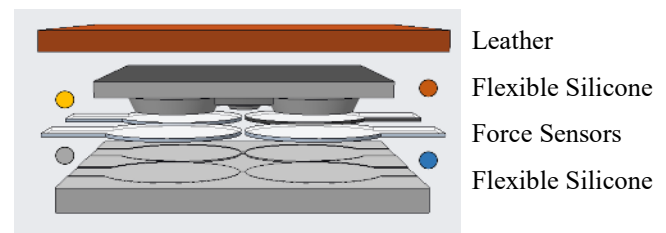


Figure 1: Proposed design of the shear sensor showing the four key layers in the design. An upper leather layer to approximate skin, two layers of flexible silicone surrounding four Force Sensitive Resistors.

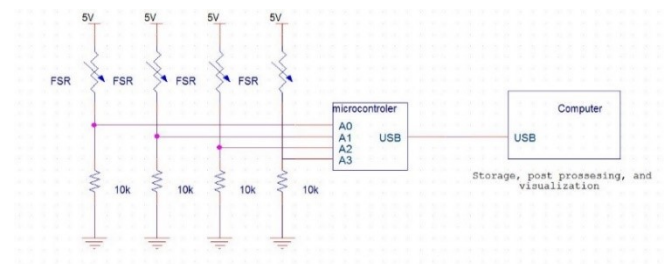


Figure 2: Basic circuit diagram of the shear sensor consisting of 4 Force Sensitive Resistors connected to an Arduino UNO and then to PC through the USB port.

B. The sensor layers

The planar structure of the shear sensor is shown in Figure 1 and an image of an actual prototype system is shown in Figure 3. The structure consists of two silicone layers, a leather layer, and an FSR layer.

The FSR layer contains four Interlink model 402 FSRs arranged in a 2x2 grid. The dots in Figure 1 correspond to the position of the FSR in the grid i.e. the yellow dot represents FSR1 followed by the brown, gray and blue as FSR4. The overall size of the sensor was approximately 4.5 cm square which is adequate for the transfer application. Smaller sensor sizes will be limited by the size of the FSR component. The prototype that was built based on the specifications given above can be seen in Figure 3.

The silicone layers are used to sandwich the FSR layer and ensure that the FSR layer is physically secure as well as

to help evenly distribute any applied normal forces to each of the FSRs [18]. The silicone layers were created using Smooth-On Ecoflex™ 00-35 gel and were created within 3D printed molds. The bottom silicone layer of the sensor was designed to hold the FSRs in their desired orientation. To accomplish this function, indents were molded into the silicone such that the FSRs are adjacent to each other and so that there is a 3 mm gap between the FSR and the edge of the silicone. The FSRs are located close to each other to minimize the amount of dead area internally within the sensor where the silicone layers can contact directly bypassing the FSRs while also ensuring the FSR devices do not mechanically interfere or touch each other.

The upper layer of silicone was designed to distribute any normal force applied to the top of each of the FSRs equally and is shown in Figure 4. This was accomplished by adding 5 raised pads to the FSR facing side of the layer. Four of the pads have the same size as the active region of the FSR and are located over the center of each FSR. These pads protrude approximately 3 mm from the body of the layer. The fifth pad is located at the center of the layer, has a radius of 2.5mm and protrudes approximately 3.5 mm above the body of the layer. This pad is used to prevent the top layer from collapsing if a force is applied directly to the center of the sensor.

To prevent the top layer from collapsing when a force is not applied directly to the center of the FSR pads two features were added. The first feature is a circular extrusion located at the center of the layer with a radius of 1.35 cm. A cross structure that connects each of the FSR pads was extruded from the top of the previously mentioned circle structure. The circular structure was added to decrease how much the top layer bends when a force is applied in between the FSR pads. The cross pattern was added to help evenly distribute normal forces applied to the sensor by increasing the connection between the FSR pads. Figure 4 shows an image of the features described above.

Leather selected for the upper layer because it is thin and closely matched human skin [18]. The leather layer was added as the top layer to give the sensor a more skin-like approximation in how the sensor interacts with transfer devices either directly or through clothing materials.



Figure 3: Prototype of the shear sensor showing the prototypes for each of the same four layers as shown in Figure 1.

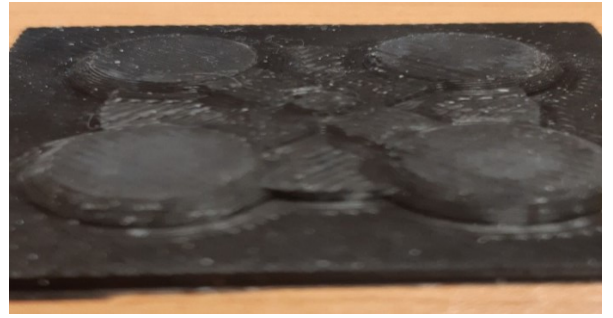


Figure 4: Image showing the detailed features on the FSR facing surface of the upper silicone layer.

C. Software

There are two pieces of software written to allow the system to function. The first is used to set up the Arduino to acquire data from the sensor and to initialize the serial port. The PC is running a MATLAB application that reads the COM port and performs the post-processing of the data.

The firmware running on the Arduino uses the 10-bit linear ADCs with a range from 0 to 5V on the analog pins of the board to sample the output voltage of the voltage divider. The program samples each FSR at 1000 Hz and averages 20 sample blocks to provide a low pass filter with a cut off frequency of approximately 22 Hz to remove high-frequency noise from the samples. The average values are then decimated by a factor of 20 leading to a 50 Hz sampling rate for each of the sensors, which is then sent to the PC via the COM port. Each time data is sent from the Arduino to the PC the packet contains the time stamp and the twenty sample block average for each of the FSRs.

Once a packet of data reaches the computer, an interrupt is triggered that halts the current task and runs a function that reads the data from the COM port. The packet of voltage data is then sent to a function that uses the R_{known} value of the lower voltage divider resistance to determine the FSR conductance in terms of Siemens. This is accomplished by rearranging (1) to solve for the FSR value. The equation is then inverted and scaled to represent the conductance value of the FSR [17]. Equation 2 is the resulting equation from the above manipulations where V_{read} is the output voltage from the voltage divider, V_{cc} is the voltage supplied to the voltage divider and R_{known} is 10 k Ω . The result of (2) is then plotted in real-time.

$$V_{read} = V_{cc} \left(\frac{R_{known}}{R_{known} + R_{FSR}} \right) \quad (1)$$

$$G_{FSR} = \frac{1}{R_{FSR}} = \frac{V_{read}}{(V_{cc} - V_{read}) * R_{known}} \quad (2)$$

III. RESULTS

The sensor was evaluated to determine how the sensor responds to forces of various types including how the sensor responds to changing normal forces in the absence of any shear. Then how the sensor responds in various cases of shear forces such as different shear forces with a constant normal

force and different normal forces with a constant shear. Lastly, drift in the sensor was evaluated for a short time interval of a few minutes that would be expected during transfer events.

A. Varying normal force without shear

To test the linearity of the prototype sensor in response to normal forces, weights were applied to the top of a flat metal disk with a diameter of 2.1 cm located at the center of the prototype. Once the output of the sensor had settled at its maximum, twenty seconds of data was collected. The force applied to the sensor ranged from 0.88 N to 4.9 N. The test was performed five times for each weight with a sixty-second rest period in between each weight application. The average value for the five measurements was computed.

Figure 5 shows the average conductance of each FSR given different applied normal forces. A first-order polynomial was fitted to each FSR data set. Each of the first order models was able to achieve an R^2 value of approximately 0.996, which shows that each of the FSRs is linear over the range of 0.85 to 4.9 N which is within the range required for human body measurement given the size of the sensor and the expected forces for the various body segments on the half-scale human model that will be initially used on lift testing.

From the first order models in Figure 5, forces less than 0.8 N the sensor (0.2 N per FSR) show a floor effect of the sensor as this level is reaching forces close to the FSR specified a minimum of 0.1 N [17]. Figure 5 also shows that the sensors begin to deviate away from each other at 4.9 N and this deviation is likely explained by variation in how the applied force was distributed through the sensor

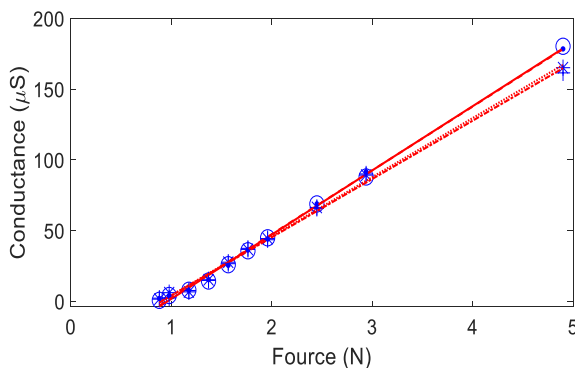


Figure 5: Experiment of linearity to weights being applied to the top of the sensor. The data points for FSR1 through FSR4 are represented by the “o”, “*”, “+”, and “.” markers respectively. The solid, dotted, dash-dot and dashed lines represent the first order best fit lines for the above data points and have R^2 values of 0.997, 0.996, 0.995, and 0.997 respectively.

B. Varying normal force with constant shear

The linearity of the system in response to a constant shear force while the normal force is varied was tested by applying weights to the top of a flat metal disk with a diameter of 2.1

cm located at the center of the prototype. The normal force applied to the sensor ranged from 0.98 N to 4.9 N in 0.49 N (50 g) steps. The shearing mass used was 70 g leading to a shear force of 0.69 N. Once the normal response stabilized, a shear force was applied using a pulley system and weight to translate the downward force of the weight due to gravity into a shear force on the sensor. Twenty seconds of data once the output of the FSR conductance values had stabilized at their peak was captured. This test was repeated five times for each normal weight used and the average value of all the measurements were taken.

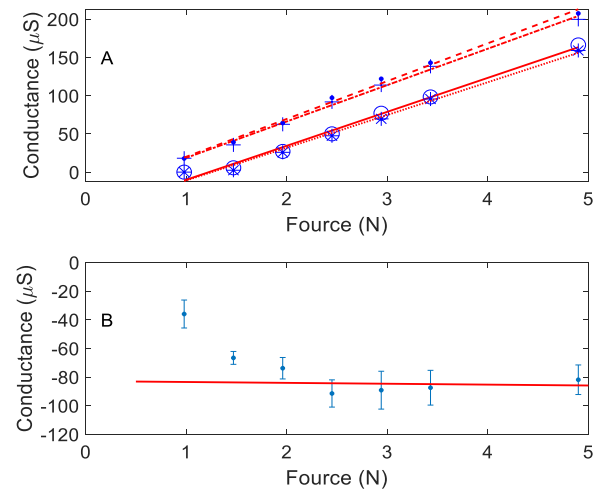


Figure 6: Determining linearity. A) increasing weights placed on the sensor with a constant 70g mass creating a shear force. The data points for FSR1 through FSR4 are represented by the “o”, “*”, “+”, and “.” markers respectively. The solid, dotted, dash-dot and dashed lines represent the first order best fit lines for the above data points and have R^2 values of 0.990, 0.987, 0.996, and 0.994 respectively B) shows the difference of the conductance for the FSRs per equation 3.

Figure 6A shows the average conductance for each of the FSRs given a constant shear force and a varying normal force. A first-order model was fitted to the data collected for each of the FSRs. Using this model, it was found that the R^2 value for each fit was between 0.985 and 0.995, this shows that the sensors remains linear for increasing normal forces given a constant shear force. Figure 6B was created using (3) as the shear force was applied perpendicular to one edge of the sensor such that if a line was drawn through the sensor in the direction of the shear force applied, FSR1 and 2 would be the two sensors experiencing increased force while FSR3 and 4 would experience reduced force.

$$diff = FSR1 + FSR2 - FSR3 - FSR4 \quad (3)$$

Examining figure 6B it appears that the difference between the FSRs reaches a constant value for larger normal forces while deviating from this at lower normal force levels. One

expected cause of the deviation is the floor effect for the FSRs that are being unloaded by the shear. This floor effect is visible in Figure 6A for the values from FSR 3 and 4 at loads of below 1.5 N. A first order polynomial was fitted to the section of figure 6B above 2 N and was found to have an R^2 value of 0.998.

C. Constant normal with varying shear

In this test, which is more typical of how the sensor will be used in lift evaluations, the normal force is held constant and the shear force is varied. To measure the shear force applied to the sensor, the difference in conductance between the two FSR pairs is needed per equation (3). To accomplish this, a 1.96 N normal force was applied to the center of a metal disk located at the center of the prototype. A shear force ranging from 0 N to 1.47 N was applied to the sensor using a series of weights attached using a pulley system to convert a downward force into a shear force. Twenty seconds of data was sampled after the output stabilized. The test was repeated five times for each shear force applied and the average of all the samples was recorded.

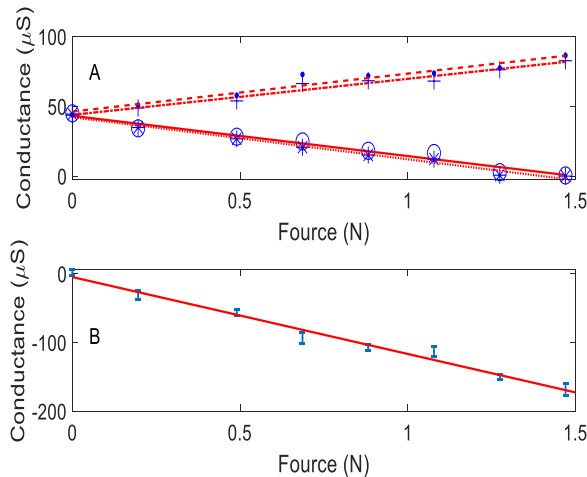


Figure 7: Linearity experiment. A) increasing the shear force applied to the sensor while keeping the normal force constant. The data points for FSR1 through FSR4 are represented by the “o”, “*”, “+”, and “.” markers respectively. The solid, dotted, dash-dot and dashed lines represent the first order best fit lines for the above data points and have R^2 values of 0.969, 0.983, 0.964, and 0.940 respectively B) shows the difference of the conductance for the FSRs per equation 3.

The average value of the conductance for each FSR given a constant normal force and varied shear force is shown in Figure 7A. First-order models were fitted to each of the data sets for the individual FSRs. Figure 7B displays the difference between the conductance of the loaded and unloaded FSR pairs for the given shear force. A first-order polynomial model was fitted to the difference. The model was found to have an R^2 value of 0.981. This indicates that the prototype is linear for the range of interest from 0 to 1.47 N.

Analyzing figure 7A the rear FSRs reach zero for forces greater than or equal to 1.47 N. This indicates that the range of the shear sensor is dependent on the amount of normal force applied to the sensor. This could be corrected by repeating the experiment with increased normal masses and record the point that the sensor bottoms out then fit a calibration curve to the data.

D. Sensor short term stability

The drift of the sensor was tested by applying a 1.96 N to the center of a metal disk located in the middle of the sensor. Figure 8A shows the raw conductance values for each of the sensors for the test. Figure 8B was created by plotting the result of (3) using the data collected. A first order polynomial was fitted to the data in figure 8B. Examining the fit in figure 8B from the slope of the fit suggests that the sensor drifts approximately 0.01 μ S per second.

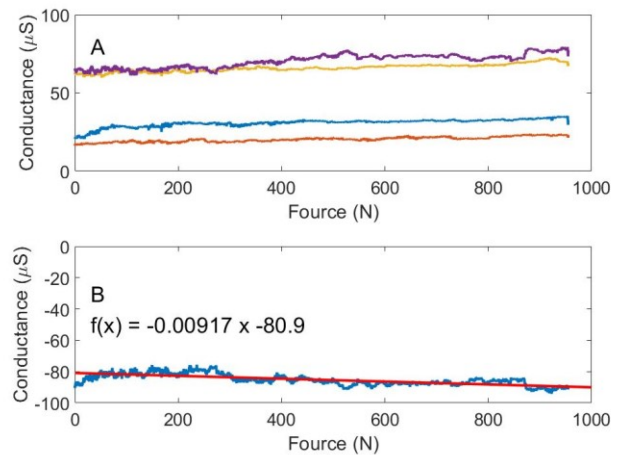


Figure 8: Drift of the sensor. A) conductance values for each of the sensors over a 16-minute period with a 200g normal mass and 70g shear mass B) shows the difference of the conductance for the FSRs per equation 3.

IV. CONCLUSION

This paper introduces a new FSR based sensor that is capable of measuring both normal and shear forces applied to skin. The sensor uses force sensitive resistors to measure the force being applied to it. Taking the aggregate of the force applied to each FSR in the sensor the amount of shear force that is applied to the sensor can be measured. The sensor was designed to be able to measure normal forces in the range of 0.1 N to 100 N. the layered structure of the sensor was implemented in such a way to approximate the texture and feel of skin. The initial results discussed in this paper indicate that the sensor has a linear response to both shear and normal forces over the range of interest required for the accurate measurement of normal and shear forces found in both a 50 percent scale human and a full-size human.

Future work that can be done in the evaluation of this sensor would be to evaluate the response of the sensor with a shear force applied to each of the sides of the sensor so that

the directionality of the sensor can be measured. The sensor response dynamics need to be analyzed such as the step and impulse response so that the frequency response is characterized. Regression equations may also need to be created to map the conductance measured from the sensor to the force applied to it as the performance of the shear measurement is dependent on the normal force load.

Once the sensor is fully evaluated, it will also need to be testing within the target application of patient transfer systems to understand its performance in this application and subsequently the forces these systems apply to the patients during transfer.

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